ADAM9-Related Retinopathy (Cone-Rod Dystrophy 9, CORD9) — Comprehensive Disease Report

> **Causal gene (stated up front):** `ADAM9` (A Disintegrin And Metalloprotease domain, family member 9; HGNC:216; NCBI Gene 8754; UniProt Q13443; chromosome 8p11.22). The disease referenced by **MONDO:0800398** is the autosomal-recessive, early-onset **cone-rod dystrophy 9 (CORD9)** caused by biallelic loss-of-function variants in **ADAM9**. Despite the serial name, it is unambiguously an *ADAM9* disorder; this report follows the MONDO ID and causal gene given.

Evidence-source legend: **[H]** human clinical, **[M]** mouse model, **[D]** dog/natural animal model, **[V]** in vitro, **[R]** review/computational.

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1. Disease Information

Overview. ADAM9-related retinopathy is an inherited, bilateral, progressive retinal dystrophy in which **cone photoreceptors are affected first and most severely, followed by rods** (a "cone-rod" pattern). It is designated **cone-rod dystrophy 9 (CORD9)**. Onset is typically in early childhood with reduced central/color vision, photophobia (day blindness) and later night blindness and peripheral field loss. The initiating lesion is at the **photoreceptor outer-segment–retinal pigment epithelium (POS–RPE) junction** rather than within the photoreceptor itself [H/M,].

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Key identifiers. - **MONDO:** MONDO:0800398 - **OMIM (phenotype):** 612775 — "Cone-rod dystrophy 9" (CORD9) - **OMIM (gene):** 602713 — ADAM9 - **HGNC:** HGNC:216 (ADAM9); **NCBI Gene:** 8754; **Ensembl:** ENSG00000168615; **UniProt:** Q13443 - **Orphanet:** within "Cone-rod

dystrophy" (ORPHA:1872); ADAM9 is a recognized causative gene - **ICD-10:** H35.5 (Hereditary retinal dystrophy); **ICD-11:** 9B70.0 (Hereditary retinal dystrophy, cone-rod) - **MeSH:** Cone-Rod Dystrophies (D000076080); Retinal Dystrophies (D058499)

Synonyms / alternative names. Cone-rod dystrophy 9; CORD9; CRD9; ADAM9-related cone-rod dystrophy; autosomal recessive cone-rod dystrophy due to ADAM9; (animal homologs: canine cone-rod dystrophy 3 / crd3; canine generalized progressive retinal atrophy, gPRA, in Glen of Imaal Terrier).

Clinical-spectrum note. Although classified as a "cone-rod dystrophy," ADAM9 disease in humans is frequently **early-onset and severe**, with some patients presenting in the first years of life with marked visual impairment/nystagmus, so that the phenotype overlaps the **early-onset severe retinal dystrophy (EOSRD) / Leber congenital amaurosis (LCA) spectrum** and ADAM9 is included on EOSRD/LCA and generic inherited-retinal-disease gene panels [H, PMIDs 25091951, 25546566, 38892339]. This is a more severe, earlier-onset presentation than the classic adult-onset cone-rod dystrophies.

Information source. Disease-level, aggregated from small consanguineous-family case series and from mouse/dog models — **not** from large EHR cohorts. The condition is rare and most publications describe individual families or single patients [H, PMIDs 19409519, 25091951, 25546566, 38028590]. Panel-sequencing cohorts of childhood retinal dystrophy (e.g., a Chilean EOSRD/LCA cohort of 67 patients/60 families) illustrate that ADAM9 is among the recurrently tested genes in this diagnostic category [H,].

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2. Etiology

Primary cause — genetic. Biallelic (homozygous or compound-heterozygous) **null/loss-of-function mutations in ADAM9**. Discovery: "By autozygosity mapping, we identified null mutations in the ADAM metallopeptidase domain 9 (ADAM9) gene in four consanguineous families with recessively inherited early-onset CRD" [H,].

P 19409519

Genetic risk factors. - *Causal variants:* nonsense, frameshift, splice-site and whole-exon/large deletions in ADAM9 (see §4). - *Consanguinity / founder homozygosity* is the dominant risk context; most reported families are consanguineous (Middle East, North Africa, South Asia) [H, PMIDs 19409519, 25091951]. - *Carrier status:* heterozygotes are unaffected (recessive). - No established common-variant susceptibility loci or modifier genes are reported.

Environmental / lifestyle risk factors. None established as causal. The disease is monogenic. (Light exposure is a generic theoretical accelerant of photoreceptor stress in retinal dystrophies but is **not** demonstrated specifically for ADAM9.)

Protective factors. None genetically defined. General retinal-health measures (UV/blue-light protection, avoidance of smoking) are prudent but unproven for this disease.

Gene–environment interactions. Not characterized for CORD9. Of note, in RPE cells *hyperglycemia upregulates ADAM9*, increasing MerTK shedding and impairing outer-segment phagocytosis [V, P 35163068] — a metabolic modulator of ADAM9 biology, though not shown to modify CORD9 severity.

3. Phenotypes

Phenotypes are **clinical signs, symptoms, and electrophysiological/laboratory (ERG, OCT) abnormalities**. Onset is usually **childhood/early-onset**; course is **progressive**. Frequencies are qualitative (small cohorts).

Phenotype	Type	Typical onset / course	Frequency	Suggested HPO
Cone-rod dystrophy (cone > rod loss)	Clinical sign / core	Childhood; progressive	Defining (~100%)	HP:0000548 (Cone/cone-rod dystrophy)
Reduced visual acuity / visual impairment	Symptom	Childhood; progressive	Very frequent	HP:0000505 / HP:0032037
Photophobia ("day blindness")	Symptom	Early	Frequent	HP:0000613
Color vision defect (dyschromatopsia)	Symptom	Early	Frequent	HP:0000551
Nyctalopia (night blindness)	Symptom	Later (rod involvement)	Common as disease advances	HP:0000662
Constricted / peripheral visual field loss	Sign	Progressive	Common	HP:0001133
Nystagmus	Sign	Early if infantile onset	Variable	HP:0000639
Abnormal / reduced ERG (photopic > scotopic)	Lab/ electrophysiology	At diagnosis	Defining	HP:0000512
RPE changes / pigmentary retinopathy, mid-peripheral greyish-white discoloration	Sign (fundus)	Progressive	Frequent	HP:0000580 / HP:0007894
Attenuated retinal vasculature	Sign	Advanced	Common	HP:0007843
Optic disc pallor	Sign	Advanced	Reported	HP:0000543
Macular atrophy / "coloboma-like" macular lesion	Sign	Variable	Reported in some	HP:0007754

Phenotype	Type	Typical onset / course	Frequency	Suggested HPO
Cataract (anterior polar / posterior subcapsular)	Sign	Variable	Reported in some families	HP:0000518

Representative description: childhood-onset visual impairment, "reorganisation of the retinal pigment epithelium with mid-periphery greyish-white discolouration, attenuated retinal vasculatur and optic disc pallor," with a coloboma-like macular lesion in one case, plus anterior polar/posterior subcapsular cataract [H,].

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Severity: generally moderate-to-severe with legal blindness common by adulthood; **variable expressivity** even within families. **Quality-of-life impact:** substantial — early loss of central acuity, color vision and glare disability impair reading, schooling, driving (precluded), and independent navigation; night blindness and field constriction add mobility impairment. No disease-specific EQ-5D/SF-36 datasets are published.

4. Genetic / Molecular Information

Causal gene. ADAM9 (OMIM 602713; HGNC:216; 8p11.22). A single-gene, autosomal-recessive disorder.

Pathogenic variants. - **Variant types:** nonsense, frameshift (small indels), canonical splice-site, and multi-exon genomic **deletions** — all converging on **loss of function/null alleles**. Examples: - Homozygous splice-acceptor **c.1396-2A>G** (exon 14 region) — consanguineous Egyptian family [H,].

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- Compound-heterozygous ADAM9 variants detected by exome sequencing in a child with CRD [H,].

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- (Animal homolog, mechanistically informative) canine **deletion of exons 15–16** → frameshift

+ premature stop, removing catalytic/downstream domains [D, PMIDs 20806078, 20691256]. - **ACMG classification:** reported disease alleles are predominantly **pathogenic / likely pathogenic** (null variants meeting PVS1 + recessive segregation + rarity). - **ClinVar spectrum (NCBI E-utilities, 2026):** 718 ADAM9 records — **101 Pathogenic, 17 Likely pathogenic, 623 VUS**. Among sequence-level P/LP variants the molecular consequences are overwhelmingly **truncating/LoF: nonsense (16), frameshift (7), splice-acceptor (6), splice-donor (6), with only 1 missense**; types include SNVs (28) and small deletions (11). Many "pathogenic" entries are large regional copy-number gains/losses spanning ADAM9 among multiple genes and are **not** CORD9-specific. The large VUS fraction and low pathogenic-missense count mean splice/functional assessment and intragenic deletion/CNV detection materially aid molecular diagnosis. *(Retrieved programmatically via ClinVar E-utilities.)* - **Allele frequency:** individual pathogenic alleles are rare/absent in gnomAD (consistent with a rare recessive disorder); specific population allele frequencies are not systematically catalogued. - **Population genetic constraint (gnomAD, GRCh38; ENSG00000168615, chr8:38,996,754–39,105,445):** ADAM9 is **not haploinsufficient — pLI = 2.4×10^{-6} (≈0), oe_lof = 0.499 (90% CI 0.40–0.63; LOEUF ≈ 0.63)**, 54 observed vs 108.1 expected LoF variants (LoF z = 4.42; missense z = 2.26, oe_mis = 0.83). Heterozygous LoF is tolerated (healthy carriers), concordant with autosomal-recessive, biallelic-null disease biology and excluding a dominant/haploinsufficiency mechanism. *(Retrieved programmatically via gnomAD GraphQL API.)* - **Origin: germline**, biallelic. (Somatic ADAM9 changes occur in cancer but are unrelated to the retinopathy.) - **Functional consequence: loss of function** of the ADAM9 metalloprotease/adhesion protein (not gain-of-function or dominant-negative), confirmed by concordant human null mutations and knockout mouse/dog phenotypes [H/M/D,].

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Modifier genes. None established; intra-familial variable expressivity suggests unidentified genetic/environmental modifiers.

Epigenetic information. No disease-specific methylation/chromatin findings reported.

Chromosomal abnormalities. Large intragenic deletions (exon-level) occur; no recurrent whole-chromosome aneuploidy or translocation is associated.

Suggested ontology: gene **HGNC:216**; GO molecular function **GO:0004222** (metalloendopeptidase activity), **GO:0008237** (metallopeptidase activity).

5. Environmental Information

- **Environmental factors:** none established as causal or necessary; the disease is fully penetrant genetically.
- **Lifestyle factors:** no proven role; general counseling on UV/blue-light protection and smoking avoidance is extrapolated, not evidence-based for ADAM9.
- **Infectious agents:** not applicable — non-infectious Mendelian disease.

(Mechanistic aside: ADAM9 expression is metabolically regulated — e.g., upregulated under high glucose in RPE [V, P 35163068] — but no environmental trigger is implicated in CORD9 onset.)

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic null ADAM9 variants → loss of functional ADAM9 metalloprotease/adhesion protein** at the apical surface of the RPE (and photoreceptor interface). *[demonstrated — human null mutations + KO models]*
2. Loss of ADAM9 → **disorganization of RPE apical processes/microvilli** and loss of normal integrin/ectodomain-shedding–mediated adhesion between RPE and photoreceptor outer segments. *[demonstrated in mouse and dog]* "the apical processes of the retinal pigment epithelium (RPE) cells are disorganized and contact between photoreceptor outer segments (POSs) and the RPE apical surface is compromised" [M, P 19409519]; "the primary lesion … appears to be a failure of the apical microvilli of the retinal pigm[ent epithelium]" [D, P 20806078].
3. Disrupted POS–RPE apposition → **expansion of the subretinal space**. A compensatory response occurs: RPE forms **elongated pseudopods** that bridge the gap to continue ingesting outer-segment tips, temporarily preserving outer-segment renewal and

photoreceptor function. *[demonstrated in mouse]* [M,

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4. When compensation fails over time → **progressive photoreceptor outer-segment shortening/disorganization and outer nuclear layer thinning**, with RPE basal deposits and macrophage infiltration. *[demonstrated in aged KO mice]* [M,

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5. Photoreceptor loss (cones preferentially, then rods) → **reduced photopic then scotopic ERG, loss of central/color vision, photophobia, then night blindness and field constriction** — the clinical cone-rod dystrophy. *[inferred from model + human concordance]*

Branch (modulatory, inferred): Because ADAM9 sheds the phagocytic receptor **MerTK** and regulates **ADAM10 α-secretase**, ADAM9 loss may also perturb outer-segment phagocytosis and local growth-factor/receptor signaling at the RPE, contributing to the renewal defect [V, PMIDs 35163068, 21956108]. This branch is mechanistically plausible but not yet demonstrated to drive CORD9.

Molecular pathways / processes. Ectodomain shedding (sheddase activity), integrin-β1–mediated cell adhesion/migration, and signaling modulation (ADAM9 influences EGFR/AKT, mTOR, NF-κB, STAT3 pathways) [R,

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physiologically relevant RPE process is **photoreceptor outer-segment phagocytosis/renewal** and POS–RPE adhesion. Suggested GO terms: **GO:0006911/GO:0043277** (phagocytosis / apoptotic cell clearance), **GO:0007155** (cell adhesion), **GO:0006508** (proteolysis), **GO:0001895** (retina homeostasis).

Protein dysfunction. ADAM9 is a type-I transmembrane multidomain metalloprotease (prodomain, catalytic Zn-metalloprotease, disintegrin, cysteine-rich, EGF-like, transmembrane, cytoplasmic). Disease alleles truncate/abolish the protein → **complete loss of catalytic and adhesion function** [R,

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Cell types / compartments. Primary: **retinal pigment epithelial cell (CL:0002586)** apical microvilli; secondary: **retinal cone (CL:0000573)** then **rod (CL:0000604)** photoreceptors. Subcellular: plasma membrane / apical microvilli, extracellular/subretinal space; (ADAM9 also at endosomes/Golgi during trafficking).

Immune/inflammation, tissue-damage. Secondary **macrophage** recruitment and **RPE basal (drusen-like) deposits** in late disease [M,]; P 19409519 chronic photoreceptor loss is degenerative, not autoimmune. Oxidative stress is a generic downstream contributor (inferred).

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7. Anatomical Structures Affected

- **Organ level — primary: eye / retina** (UBERON:0000970 eye; UBERON:0000966 retina). **Body system:** visual/nervous system.
- **Secondary: lens** (cataract in some families; UBERON:0000965), optic nerve head (pallor; UBERON:0000941), retinal vasculature (attenuation; UBERON:0003516).
- **Tissue/cell level: retinal pigment epithelium** (UBERON:0001782; cell CL:0002586) — the initiating site; **neural retina photoreceptor layer** — cones (CL:0000573) then rods (CL:0000604); outer nuclear layer thinning.
- **Subcellular:** RPE **apical microvilli / processes** and **plasma membrane**; **subretinal space** (POS–RPE interface). Suggested GO cellular component: **GO:0005886** (plasma membrane), **GO:0005902** (microvillus).
- **Localization / laterality: bilateral**, generally symmetric; disease is **posterior-segment** centered with early macular/central and later mid-peripheral involvement.

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8. Temporal Development

- **Onset: early childhood** ("early-onset CRD") in humans [H,]; P 19409519 childhood-onset visual impairment typical [H,

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Pattern is **insidious/chronic**, not acute.

- **Progression:** slowly **progressive** over years to decades; **cone-first** functional loss (acuity, color, photophobia) preceding **rod** involvement (night blindness, field loss). Variable rate and expressivity.
- **Stages:** early (photophobia, reduced acuity, abnormal photopic ERG, subtle RPE changes) → intermediate (color/field loss, pigmentary RPE changes, vascular attenuation) → advanced (macular/widespread atrophy, severe visual loss / legal blindness, optic disc pallor).
- **Course / duration: chronic, lifelong**; no spontaneous remission. (Animal models are informative on tempo: mouse is comparatively mild/late — photoreceptors near-normal at 12 months, degeneration by 20 months [M,

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dog disease is adult/late-onset with variable, at times very late onset [D,

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- **Critical window for intervention:** the period of preserved photoreceptors with compensatory RPE pseudopods [M,

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defines a **therapeutic window** before irreversible photoreceptor loss (inferred).

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9. Inheritance and Population

- **Inheritance: autosomal recessive** (AR) [H,

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- **Penetrance:** essentially **complete** in biallelic carriers (age-dependent emergence of signs).
- **Expressivity: variable**, including intra-familial variability (e.g., associated cataract in some members) [H,

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- **Carrier frequency:** heterozygous carriers unaffected; pathogenic alleles individually rare in gnomAD. Specific carrier-frequency estimates are not established, but gnomAD

constraint confirms heterozygous LoF is tolerated (**pLI ≈ 0, LOEUF ≈ 0.63**; 54 obs vs 108 exp LoF variants), consistent with a healthy-carrier recessive model.

- **Consanguinity / founder effects:** strongly associated with **consanguineous pedigrees** (autozygosity); reported families from Egypt, Middle East, South Asia and the UK [H, PMIDs 19409519, 25091951, 25546566]. No single global founder allele; private family mutations predominate.
- **Anticipation / mosaicism:** not applicable (no repeat expansion); germline mosaicism not reported.
- **Epidemiology: rare.** No disease-specific prevalence/incidence is published. For context, all cone-rod dystrophies together have an estimated prevalence of ~1 in 30,000–40,000, and ADAM9 is a **minor** contributor overall but enriched in consanguineous populations. Both sexes affected equally (**M:F ≈ 1:1**), consistent with autosomal inheritance.

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10. Diagnostics

Clinical / functional tests. - **Full-field electroretinography (ERG):** reduced responses with **photopic (cone) dysfunction ≥ scotopic (rod)** — hallmark of cone-rod pattern (HPO HP:0000512) [H/M, P 19409519]. - **Optical coherence tomography (OCT):** outer-retinal/ellipsoid-zone disruption, ONL thinning, RPE changes, subretinal-space abnormalities. - **Fundus examination / autofluorescence:** RPE reorganization, mid-peripheral greyish-white discoloration, vascular attenuation, optic disc pallor, macular atrophy [H, P 25091951]. - **Color vision and visual-field testing:** early dyschromatopsia; progressive field constriction. - Biopsy/histopathology is not used clinically (findings known from models: disorganized RPE apical microvilli, POS–RPE separation, basal deposits, macrophages).

Genetic testing (definitive). - Approach: **molecular genetic testing of ADAM9** via **inherited-retinal-disease gene panels** or **whole-exome/whole-genome sequencing**; single-gene/Sanger confirmation of candidate variants; **CMA/deletion analysis** to detect multi-exon deletions (important given large-deletion alleles). Exome sequencing has directly diagnosed ADAM9 CRD in children [H,

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Targeted enrichment panels (e.g., IROme) plus Sanger confirmation were used historically [H,

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Because the ClinVar P/LP spectrum is truncating/splice-dominated with a large VUS fraction (623 VUS) and few pathogenic missense, **including copy-number/intragenic-deletion analysis and RNA/splice assays improves diagnostic yield**. - Karyotype/FISH/mtDNA/repeat-expansion testing: not indicated.

Clinical criteria / differential diagnosis. Diagnosis = compatible cone-rod ERG/OCT phenotype + biallelic pathogenic ADAM9 variants. **Differentials:** other cone-rod dystrophies (ABCA4, CRX, GUCY2D, RPGR, RIMS1, PROM1, etc.), achromatopsia (CNGB3/CNGA3), Leber congenital amaurosis, and Stargardt disease — distinguished by gene testing and phenotype nuances.

Screening. Cascade/carrier testing within affected families; preconception/prenatal testing possible given known familial variants. No population newborn screening.

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11. Outcome / Prognosis

- **Survival/mortality: normal life expectancy**; ADAM9 retinopathy is **not life-limiting** — it is an isolated (non-syndromic) ocular disorder.
- **Morbidity / disability: progressive visual disability** is the principal burden — early loss of central acuity, color vision and glare tolerance, later night blindness and peripheral field loss, frequently culminating in **legal blindness** in adulthood. Major impact on education, employment, mobility and independence (ICF visual-function domains).
- **Disease course / complications:** chronic progressive vision loss; cataract in some patients (surgically treatable); no systemic complications.
- **Recovery potential:** none spontaneously; vision loss is irreversible once photoreceptors degenerate.
- **Prognostic factors:** earlier onset and more complete null genotypes tend to associate with more severe course (inferred); residual photoreceptor/ellipsoid-zone integrity on OCT is a practical prognostic/therapeutic-window biomarker.

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12. Treatment

No approved disease-specific or curative therapy currently exists. Management is **supportive and rehabilitative**: - **Low-vision rehabilitation** (NCIT: low-vision aids), magnification, high-contrast tools, orientation/mobility training. - **Photophobia management:** tinted/edge-filter lenses, sunglasses, environmental lighting adaptation. - **Cataract surgery** when visually significant cataract is present [H, P 25091951]. - **Genetic counseling** (see §13). - **General retinal support:** correction of refractive error; UV protection (prudential).

Experimental / future directions. - **Gene-replacement (AAV-mediated ADAM9 augmentation)** is a biologically rational strategy — ADAM9 is a small, recessive loss-of-function gene expressed in RPE/photoreceptors, matching the paradigm of RPE-directed retinal gene therapy; the preserved-photoreceptor/pseudopod window [M, P 41642651] supports a treatable interval. (Status: **preclinical/investigational**; no approved product. No specific completed human trial identified in this review.) - Other modalities under general IRD research (optogenetics, cell replacement) are not ADAM9-specific.

Pharmacogenomics: not applicable. **Treatment outcomes / adverse events:** none established for disease-specific therapy.

Suggested NCIT concepts: Gene Therapy, Adeno-Associated Viral Vector, Supportive Care, Cataract Surgery, Low Vision Aid.

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13. Prevention

- **Primary prevention:** not possible for a monogenic recessive disease except via **reproductive genetic counseling**. In consanguineous communities, awareness and **carrier screening** can reduce incidence.
- **Secondary prevention:** early diagnosis (gene testing in children with photophobia/reduced acuity/abnormal cone ERG) to enable timely low-vision support and candidacy for future therapies.

- **Tertiary prevention:** manage complications (cataract surgery), maximize residual vision via rehabilitation.
- **Genetic counseling:** recurrence risk 25% for offspring of two carriers; **cascade carrier testing**, **preimplantation genetic testing (PGT)** and **prenatal diagnosis** available where familial variants are known. Counseling on consanguinity-related risk is central.
- **Public-health / behavioral / immunization measures:** not applicable.

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14. Other Species / Natural Disease

- **Taxonomy / natural disease:** Naturally occurring ADAM9 retinopathy is well documented in the **dog (*Canis lupus familiaris*, NCBI Taxon 9615)** — **canine cone-rod dystrophy 3 (crd3)** and **generalized progressive retinal atrophy (gPRA)** in the **Glen of Imaal Terrier** breed (VBO/breed-specific), caused by an **ADAM9 exon 15–16 deletion** [D, PMIDs 20806078, 20691256] (OMIA entry for ADAM9/cone-rod dystrophy in dog).
- **Orthologous genes:** mouse *Adam9* (NCBI Gene 11491), dog *ADAM9* — functionally conserved.
- **Comparative biology:** across human, mouse and dog the **primary lesion is the RPE apical microvilli / POS–RPE junction**, confirming deep evolutionary conservation of ADAM9's role in retinal homeostasis; canine disease is a validated large-animal model "with superior potential to elucidate functional consequences of the … human ADAM9 null mutations" [D, P 20691256].
- **Zoonotic potential:** none (genetic disease).

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15. Model Organisms

- **Mouse (*Mus musculus*): Adam9-knockout** mouse — reduced photoreceptor ERG responses; at 12 months RPE apical-process disorganization with normal photoreceptors; by 20 months progressive degeneration (ONL thinning, POS disorganization, RPE basal deposits, macrophages) [M, P 19409519].

Recent work shows **compensatory RPE pseudopods** maintaining outer-segment renewal before degeneration [M,

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Strengths: genetically defined null matching human LOF; dissects RPE-interface mechanism. **Limitation:** milder/later phenotype than human (species difference in tempo). Resource: MGI (Adam9).

- **Dog (Glen of Imaal Terrier):** naturally occurring ADAM9-null crd3/gPRA — adult/late-onset, variable; valuable **large-eye preclinical/therapeutic model** [D, PMIDs 20806078, 20691256]. Resource: OMIA.
- **Model types available:** germline knockout (mouse); spontaneous null (dog). In-vitro **RPE cell models (e.g., ARPE-19)** are used to study ADAM9 sheddase biology (MerTK shedding) [V,

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- **Phenotype recapitulation:** both mouse and dog reproduce the **core RPE apical-microvilli / POS–RPE junction lesion** and progressive photoreceptor loss; they under-represent the **early, severe cone-dominated childhood onset** seen in humans.

Evidence Summary (key PMIDs)

PMID	Type	Contribution
19409519	H/M	Landmark: ADAM9 null mutations cause human early-onset CRD; KO mouse; POS–RPE junction lesion
25091951	H	Novel homozygous splice variant c.1396-2A>G; fundus phenotype; cataract association
25546566	H	Exome sequencing diagnoses ADAM9 CRD in a child
38028590	H	ADAM9 among genes in a multi-family cone-rod dystrophy cohort
20806078	D	Canine crd3 (Glen of Imaal Terrier); establishes homology with human CORD9; RPE microvilli failure
20691256	D	Canine gPRA ADAM9 exon 15–16 deletion; model for human disease
41642651	M	RPE pseudopods maintain outer-segment renewal despite subretinal space expansion
42171934	R	ADAM9 biology: domains, isoforms, substrates, signaling pathways
35163068	V	ADAM9 sheds MerTK in RPE; metabolic regulation (high glucose)
21956108	V	ADAM9 regulates ADAM10 α-secretase activity
17704059	V	ADAM9 disintegrin-cysteine-rich domain binds β1 integrin (adhesion/motility)

Database-derived quantitative evidence (this investigation)

Source	Metric	Value	Implication
gnomAD (GRCh38)	pLI	2.4×10^{-6} (≈0)	Not haploinsufficient → recessive
gnomAD	oe_lof / LOEUF	0.50 / ≈0.63 (obs 54 vs exp 108 LoF)	Under purifying selection but LoF tolerated in heterozygotes
gnomAD	missense z / oe_mis	2.26 / 0.83	Mild missense constraint
ClinVar	Total records	718	—
ClinVar	Pathogenic / Likely-path / VUS	101 / 17 / 623	Large VUS burden; interpretation challenge
ClinVar	P/LP sequence consequences	nonsense 16, frameshift 7, splice 12, missense 1	Truncating/LoF-dominated disease spectrum

Convergent conclusion: gnomAD constraint (healthy heterozygous carriers) and the ClinVar truncating-allele spectrum independently confirm that CORD9 is an **autosomal-recessive, biallelic loss-of-function** disorder — not dominant/haploinsufficient and not gain-of-function.

Key Findings & Hypothesis Status

Supported (evidence-backed): 1. ADAM9 biallelic null mutations cause autosomal-recessive early-onset cone-rod dystrophy 9 (CORD9) [H, P 19409519]. 2. The initiating lesion is at the RPE apical microvilli / POS–RPE junction, upstream of photoreceptor death [M/D, PMIDs 19409519, 20806078; compensatory RPE pseudopods, P 41642651]. 3. ADAM9 is a sheddase/adhesion metalloprotease (MerTK, ADAM10, β1-integrin) whose loss disrupts the RPE–photoreceptor interface [R/V, PMIDs 42171934, 35163068, 21956108, 17704059]. 4. Disease is typically early-onset/severe, overlapping the EOSRD/LCA spectrum [H, PMIDs 25091951, 25546566, 38892339]. 5. gnomAD constraint shows ADAM9 is NOT haploinsufficient (pLI≈0, LOEUF≈0.63) → recessive, healthy-carrier model. 6. ClinVar P/LP

spectrum is truncating/LoF-dominated (nonsense/frameshift/splice ≫ missense), with a large VUS burden. 7. Cross-species mouse and dog (Glen of Imaal Terrier) models recapitulate the RPE-interface lesion [M/D, PMIDs 19409519, 20691256].

Refuted / excluded: dominant or gain-of-function mechanism (excluded by pLI≈0 + recessive segregation + null-allele spectrum); syndromic/systemic involvement (disease is isolated/non-syndromic, normal lifespan); infectious/environmental causation (none).

Limitations and Future Directions

- The literature base is **small** (few consanguineous families, individual patients), so **phenotype frequencies, prevalence, penetrance and genotype–phenotype correlations are approximate**. No large natural-history or QoL dataset exists.
- Mechanistic depth comes largely from **mouse/dog models**, which are milder/later-onset than human disease; direct human RPE/photoreceptor molecular data are limited.
- **Gene therapy** is biologically well-justified but remains **preclinical/investigational**; no specific completed human trial was identified here — a priority for future work.
- Several ontology term suggestions (HPO/GO/CL/UBERON) should be verified against current ontology releases before database ingestion.

One-paragraph answer

ADAM9-Related Retinopathy (MONDO:0800398) is **cone-rod dystrophy 9 (CORD9)**, an autosomal-recessive, early-childhood-onset, progressive retinal dystrophy caused by **biallelic null (loss-of-function) mutations in the metalloprotease gene ADAM9**. Loss of ADAM9 disrupts the apical microvilli of the retinal pigment epithelium and the photoreceptor-outer-segment–RPE junction — the initiating lesion — leading to subretinal-space expansion and, over time, cone-predominant then rod photoreceptor degeneration with reduced ERG, photophobia, color-vision and acuity loss, and later night blindness and field constriction, culminating in severe low vision/legal blindness but normal lifespan.
